

SHOPEZ : E-Commerce Application

Full Project Documentation

1. Introduction

Project Title: SHOPEZ : E-Commerce Application

Team Members:

Ballari Vinuthna – Full Stack Developer (Individual Project). Responsible for frontend development, backend development, database design, API integration, authentication, testing, debugging, and documentation preparation.

SHOPEZ is a full-stack MERN application that provides an online shopping platform where users can browse products, view product details, add items to cart, place orders, and manage their accounts through a secure and user-friendly interface.

2. Project Overview

Purpose: The purpose of SHOPEZ is to provide users with a convenient and efficient online shopping experience while demonstrating practical implementation of MERN-stack technologies.

Features: User registration, login, product browsing, product details, shopping cart, order placement, order history, profile management, and administrative controls.

3. Architecture

Frontend: Developed using React.js and Vite with reusable components and routing.

Backend: Developed using Node.js and Express.js to provide REST APIs and business logic.

Database: MongoDB stores users, products, carts, and orders using Mongoose for database interaction.

4. Setup Instructions

Prerequisites: Node.js, MongoDB, Git, and Visual Studio Code.

Installation: Clone the repository, install dependencies in client and server folders, configure environment variables, start MongoDB, and run both frontend and backend servers.

5. Folder Structure

Client: Contains React components, pages, assets, services, and routing configurations.

Server: Contains models, controllers, routes, middleware, configuration files, and server scripts.

6. Running the Application

Frontend: Run `npm start` or `npm run dev` in the client directory.

Backend: Run `npm start` in the server directory.

After both servers are running, access the application through a web browser.

7. API Documentation

Authentication APIs handle registration and login. Product APIs retrieve product information. Cart APIs manage cart operations. Order APIs process purchases and retrieve order history.

Example Endpoints:

POST /api/auth/register

POST /api/auth/login

GET /api/products

POST /api/orders

8. Authentication

Authentication and Authorization: Implemented using JWT and bcrypt password hashing.

Tokens: JWT tokens are generated after successful login and used for protected routes.

Authorization: Only authenticated users can access account-specific features.

9. User Interface

The application includes Home, Login, Registration, Products, Product Details, Cart, Orders, Profile, and Admin Dashboard pages. The interface is responsive and user-friendly.

10. Testing

Testing Strategy: Functional and API testing were performed during development.

Testing Tools Used: Postman, Browser Testing, and MongoDB Compass.

Tested Modules: Registration, Login, Products, Cart, Orders, Profile, and APIs.

11. Screenshots or Demo

Screenshots should be included for Home Page, Login Page, Products Page, Product Details Page, Cart Page, Orders Page, Profile Page, and Admin Dashboard. A GitHub repository or deployment link can be added.

12. Known Issues

Payment gateway integration is not implemented. Product recommendation features are unavailable. Email notification services are not included. Advanced analytics can be improved.

13. Future Enhancements

Future improvements include payment gateway integration, wishlist functionality, product reviews and ratings, email notifications, mobile application support, AI-based recommendations, inventory tracking, and analytics dashboards.

Conclusion

SHOPEZ : E-Commerce Application is a full-stack MERN application developed entirely by Ballari Vinuthna. The project demonstrates React.js, Node.js, Express.js, MongoDB, authentication, API integration, cart management, order processing, and modern web development practices through a secure and scalable online shopping platform.